Accuracy in numbers :

We have two types of numbers (i) Exact (ii) Approximate

The numbers of the type $5, 8, \frac{1}{2}, \frac{3}{4}, 5.75$ will be in category of exact number, but the numbers $\frac{1}{3} = 0.3333..$, $\pi = 3.141592$ do not have finite decimal expansion and hence they are approximated to some finite digits called as significant digits for the purpose of calculations.

Sometimes we have numbers with a large number of digits such as 2.3841742 which can be rounded off to a significant figure 2.384. This process is called a rounding off the numbers.

Errors in Numerical Computations :

We have following types of errors in Numerical Calculations :

1. **Inherent Error :**
Error which exists in the problem due to approximate given data, limitations of computing aids, are called as inherent errors. The inherent error can be minimised by taking better data and using high precision computing aids.

2. **Round-off errors :** These errors are due to rounding off the numbers while process of computations. The round-off errors can be reduced by doing the computations

to more significant digits than given in data at each state of computation.

3) Truncation Errors :

These errors arises due to use of approximate formula in computation or using finite series by truncating the infinite series to some approximate terms. The study of this types of error is usually associated with the problem of convergence of infinite series.

4) Absolute, Relative and percentage errors :

If x is true value of a quantity and x' is its approximate value, then

(a) Absolute error $E_A = |x - x'|$

(b) Relative Error $E_R = \left|\frac{x - x'}{x}\right|$

(c) percentage error $E_p = \left|\frac{x - x'}{x}\right| \times 100$

Solution of Algebraic and Transcendental Equations:

(a) Algebraic equation of nth degree has the form
$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0 = 0$$

Ex: $x^2 - 2x + 1 = 0$

(b) Transcendental Equation: Non algebraic equation is called as transcendental equation.

e.g. $f(x) = \cos x - 3x + 1 = 0$

$f(x) = x e^x - 2 = 0$

$f(x) = x \log_{10} x - 12 = 0$

1. Bisection Method:

If f is a continuous function such that $f(a) < 0$ and $f(b) \geq 0$ i.e $f(a) \cdot f(b) < 0$ then $\exists\ \alpha \in [a, b]$ such that $f(\alpha) = 0$

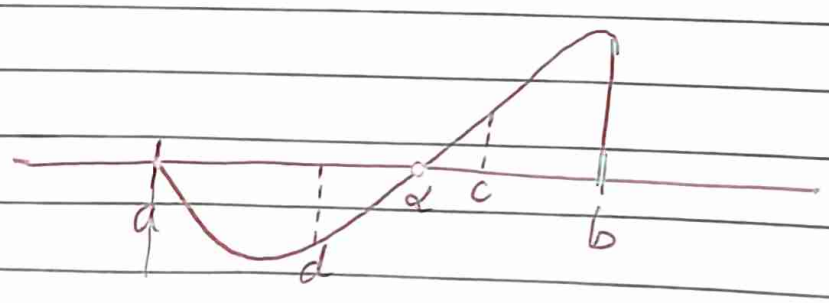
$$c = \frac{a+b}{2} \Rightarrow f(c) > 0$$

$\therefore$ Root lies between a & c

Now, $d = \dfrac{a+c}{2} \Rightarrow f(d) < 0$

$\therefore$ root lies between c and d.

Again bisect c and d and so on...



Ex.1 Apply Bisection method to find the ~~soln~~ roots / zero of function $f(x) = x^4 - x^3 - x^2 - 4$ in the interval $[1, 9]$.

Soln: $f(x) = x^4 - x^3 - x^2 - 4$ in $[1, 9]$

$f(1) = 1^4 - 1^3 - 1^2 - 4 = 1 - 1 - 1 - 4 = -5$ (-ve)

$f(9) = 9^4 - 9^3 - 9^2 - 4 = 6561 - 243 - 81 - 4 > 0$ (+ve).

$f(1) \cdot f(9) < 0$

$\therefore$ Root lies between 1 and 9.

$$C_1 = \frac{1+9}{2} = \frac{10}{2} = 5$$

$f(C_1) = 5^4 - 5^3 - 5^2 - 4 = 625 - 125 - 25 - 4 > 0$ (+ve)

$\therefore$ Root lies between 1 and 5

$$C_2 = \frac{1+5}{2} = \frac{6}{2} = 3$$

$f(C_2) = f(3) = 3^4 - 3^3 - 3^2 - 4 = 81 - 27 - 9 - 4 > 0$

Now root lies between 1 and 3.

$$C_3 = \frac{1+3}{2} = \frac{4}{2} = 2$$

$f(2) = 2^4 - 2^3 - 2^2 - 4 = 16 - 8 - 4 - 4 = 0$

Ex.2. The root of the equation lies $xe^x = 1$ lies bet^n 0 & 1 obtained by using two iterations bisection method is — (a) 0.25 (b) 0.50 (c) 0.75 (d) 0.65

Soln: $f(x) = xe^x - 1$; $f(0) = 0 - 1 = -1$ & $f(1) = e - 1 > 0$

$$C_1 = \frac{0+1}{2} = 0.5 \Rightarrow f(0.5) = 0.5\, e^{0.5} - 1$$

$$= 0.8243 - 1 = -0.1756 < 0$$

$\therefore$ Root lies bet^n 1 & $0.5 = 1/2$

$$C_2 = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4} = 0.75$$

Q.1. Find real roots of the equation $x^3 - 2x - 5 = 0$ using Bisection Method in the interval [2 3] upto three iterations.

Soln: $f(2) = 2^3 - 2(2) - 5 = 8 - 4 - 5 = -1$ (-ve)

$f(3) = 3^3 - 2(3) - 5 = 27 - 6 - 5 = 16$ (+ve)

Iteration	x_1	x_2	x_3
a (-ve)	2	2	2
b (+ve)	3	2.5	2.25
Formula	$\frac{2+3}{2} = \frac{5}{2} = 2.5$	$\frac{2+2.5}{2} = 2.25$	$\frac{2.25+2}{2} = 2.125$
$f(x)$	$f(2.5) = (2.5)^3 - 2(2.5) - 5$ $= 15.625 - 5 - 5$ $= 5.625$ (+ve)	$f(2.25) = 1.8906$ (+ve)	$f(2.125) = 0.3457$

2. Newton - Raphson Method or Successive Substitution Method :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad , n = 0, 1, 2, \ldots$$

Ex. 1. Find the root of the equation $x^4 - x - 9 = 0$ by Newton - Raphson Method, upto 4th iteration

Soln: Let $f(x) = x^4 - x - 9$

$f(1) = 1^4 - 1 - 9 = -9$ (-ve)

$f(2) = 2^4 - 2 - 9 = 16 - 11 = 5$ (+ve)

$\therefore$ $f(1)$ & $f(2)$ has opposite signs the root lies between 1 & 2.

Furthur, $f(2) < f(1)$ numerically,

$\therefore$ we take first approximation $x_1 = 2$

We have, $f'(x) = 4x^3 - 1$

Then the second approximation is given by

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2 - \frac{f(2)}{f'(2)} = \frac{2 - 5}{4 \cdot 2^3 - 1} = -2$$

$$= 2 - \frac{5}{4 \cdot 2^3 - 1} = 2 - \frac{5}{32 - 1}$$

$$= 2 - \frac{5}{31}$$

$$= 1.8387$$

$\therefore$ $\boxed{x_2 = 1.839}$

Similarly,

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.839 - \frac{[(1.839)^4 - (1.839) - 9]}{4 \cdot (1.839)^3 - 1}$$

$$\boxed{x_3 = 1.814}$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.814 - \frac{[(1.814)^4 - (1.814) - 9]}{4(1.814)^3 - 1}$$

$$x_4 = 1.81338$$

$\therefore$ The root of the given eqn is $x = 1.813$

Ex 2. Find by Newton-Raphson method, the real root of $3x - \cos x - 1 = 0$

Soln: let, $f(x) = 3x - \cos x - 1$

$f'(x) = 3 + \sin x$

Now, $f(0.6) = 3(0.6) - \cos(0.6) - 1 = 1.8 - 0.8253356 - 1$
$= -0.0253$

& $f(0.7) = 3(0.7) - \cos(0.7) - 1 = +0.3351578$

$\because$ $f(0.6)$ & $f(0.7)$ has opposite signs.

$\therefore$ The root lies between 0.6 & 0.7

Further $f(0.6) < f(0.7)$ numerically,

$\therefore$ The root lies nearer to 0.6

Let the first approximation be $x_1 = 0.61$

Let, x_2 be the second approximation.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.61 - \frac{f(0.61)}{f'(0.61)}$$

$$= 0.61 - \frac{[3(0.61) - \cos(0.61) - 1]}{3 + \sin(0.61)}$$

$$= 0.6071$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.6071 - \frac{[3(0.6071) - \cos(0.6071) - 1]}{3 + \sin(0.6071)}$$

$$= 0.607101$$

$\therefore$ The root of the given eqn is $x = 0.6071$

Q 2. Find the roots of the equation $x^4 - x - 10 = 0$ using Newton-Raphson Method upto three iterations

Soln: $f(0) = 0^4 - 0 - 10 = -10$ (-ve) | $f'(x) = 4x^3 - 1$

$f(1) = 1^4 - 2(1) - 5 = 1 - 2 - 5 = 1 - 7 = -6$ (-ve)

$f(2) = 2^4 - 2 - 10 = 16 - 12 = 4$ (+ve).

Root lies betn 1 & 2

By using Newton-Raphson Method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \& \quad x_0 = 1.5$$

put $n=0$,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.5 - \frac{f(1.5)}{f'(1.5)} = 2.015$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2.015 - \frac{f(2.015)}{f'(2.015)}$$

$$= 1.87409$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.87409 - \frac{f(1.87409)}{f'(1.87409)}$$

$$x_3 = 1.85586$$

Solution of linear Simultaneous Equations :

1. Gauss Elimination Method :

Ex.1. Solve by Gauss Elimination method

$$2x + 3y + z = 13$$
$$x - y - 2z = -1$$
$$3x + y + 4z = 15$$

pivot

Soln: we write the system as,

$$x - y - 2z = -1$$
$$2x + 3y + z = 13$$
$$3x + y + 4z = 15$$

$$AX = B$$

$$\begin{bmatrix} 1 & -1 & -2 \\ 2 & 3 & 1 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 13 \\ 15 \end{bmatrix}$$

Augmented matrix is $C = [A : B]$

$$C = \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 2 & 3 & 1 & 13 \\ 3 & 1 & 4 & 15 \end{array}\right]$$

$R_2 \longrightarrow R_2 - 2R_1$

$R_3 \longrightarrow R_3 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 5 & 5 & 15 \\ 0 & 4 & 10 & 18 \end{array}\right]$$

$R_2 \longrightarrow \dfrac{R_2}{5}$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 4 & 10 & 18 \end{array}\right]$$

$R_3 \longrightarrow R_3 - 4R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & -2 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 6 & 6 \end{array}\right]$$

$$\begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 6 \end{bmatrix}$$

$$x - y - 2z = -1$$

$$y + z = 3$$

$$6z = 6$$

$$\boxed{z = 1}$$

$$y = 3 - z = 3 - 1$$

$$\boxed{y = 2}$$

$$x = -1 + y + 2z = -1 + 2 + 2 = 3$$

$$\boxed{x = 3}$$

3. Gauss - Seidal Iteration Method :

Consider the system of equations

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

This method of iteration is applicable only for the system, where $|a_1| \geq |b_1| + |c_1|$

$|b_2| \geq |a_2| + |c_2|$ &

$|c_3| \geq |a_3| + |b_3|$

i.e in each equation the coefficient of the diagonal terms are large

i.e the system is diagonal system.

We write,

$$x = \frac{1}{a_1}\left[d_1 - b_1y - c_1z\right] \longrightarrow (1)$$

$$y = \frac{1}{b_2}\left[d_2 - a_2x - c_2z\right] \longrightarrow (2)$$

$$z = \frac{1}{c_3}\left[d_3 - a_3x - b_3y\right] \longrightarrow (3)$$

We start with initial approximation $x_0 = 0, y_0 = 0, z_0$

putting, $y = y_0$, $z = z_0$ in eqn (1)

we get $x = \frac{d_1}{a_1} = x_1$

putting, $x = x_1$ & $z = z_0$ in eqn (2)

we get y_1

putting $x = x_1$ & $y = y_1$ in (3) we get z_1

$\therefore$ x_1, y_1, z_1 is the first approximation to the soln of given system. For second iteration x_2, y_2, z_2 we use x_1, y_1, z_1 as initial values. Continue this process till the values of x, y, z are found to be desired degree of accuracy.

Ex. 1. Apply Gauss Seidal Method to solve the equations.

$$2x - 3y + 20z = 25, \quad 20x + y - 2z = 17$$
$$\& \quad 3x + 20y - z = -18$$

Soln: Rearranging the equations in proper order, we get

$$20x + y - 2z = 17$$
$$3x + 20y - z = -18$$
$$3x - 3y + 20z = 25$$

$$x = \frac{1}{20}[17 - y + 2z] \longrightarrow (1)$$

$$y = \frac{1}{20}[-18 - 3x + z] \longrightarrow (2)$$

$$z = \frac{1}{20}[25 - 2x + 3y] \longrightarrow (3)$$

we start with $y_0 = 0$, $z_0 = 0$

eqn (1) $\Rightarrow$ $x = \frac{17}{20} = 0.85 = x_1$

Using $x_1 = 0.85$, $z_0 = 0$, we get

eqn (2) $\Rightarrow$ $y = \frac{1}{20}[-18 - 3(0.85) + 0]$

$$= \frac{1}{20}[-18 - 2.55]$$

$$= \frac{-20.55}{20}$$

$$= -1.0275$$

$$= y_1$$

Using $x_1 = 0.85$, $y_1 = -1.0275$

eqn (3) $\Rightarrow$ $z = \frac{1}{20}[25 - 2(0.85) + 3(-1.0275)]$

$$= \frac{1}{20}[25 - 1.70 - 3.0825] = \frac{20.2175}{20}$$

$$= 1.0109 = \cancel{0.9683}$$

For second iteration we have initial values

$x_1 = 0.85$, $y_1 = -1.0275$, $z_1 = 1.0109$

$$x_2 = \frac{1}{20}[17 - y_1 + 2z_1] = \frac{1}{20}[17 + 1.0275 + 2(1.0109)]$$

$$= \frac{1}{20}[20.0493]$$

$$x_2 = 1.0025$$

$$y_2 = \frac{1}{20}[-18 - 3x_2 + z_1] = \frac{1}{20}[-18 - 3(1.0025) + 1.0109]$$

$$= \frac{-19.9966}{20}$$

$$= -0.9998$$

$$z_2 = \frac{1}{20}[25 - 2x_2 + 3y_2] = \frac{1}{20}[25 - 2(1.0025) + 3(-0.9998)]$$

$$= \cancel{\frac{20.001}{20}} \quad \frac{19.9956}{20}$$

$$= 0.9998$$

For third iteration we have initial values.

$x_2 = 1.0025$, $y_2 = -0.9998$, $z_2 = 0.9998$

$$x_3 = \frac{1}{20}[17 - y_2 + 2z_2] = \frac{1}{20}[17 + 0.9998 + 2(0.9998)]$$

$$= \frac{19.9994}{20} = 0.99997 \approx 1$$

$$y_3 = \frac{1}{20}[-18 - 3x_3 + z_2] = \frac{1}{20}[-18 - 3(1.0025) + 0.9998]$$

$$= \frac{-20.0077}{20} = -1.0004 \approx -1$$

$$z_3 = \frac{1}{20}[25 - 2x_3 + 3y_3] = \frac{1}{20}[25 - 2(1) + 3(-1)] = \frac{1}{20}[25-5]$$

$$= 1$$

$\therefore$ Solution of the system is $1, -1, 1$

Q.3 $6x + y + z = 105$

$4x + 8y + 3z = 155$

$5x + 4y - 10z = 65$

Sol^n:

$$x = \frac{105 - y - z}{6}$$

$$y = \frac{155 - 4x - 3z}{8}$$

$$z = \frac{65 - 5x - 4y}{-10}$$

First Iteration: $y_0 = 0$, $z_0 = 0$

$$x_1 = \frac{105}{6} = 17.5$$

$$y_2 = \frac{105 - 4x_1 - 3z_0}{8} = \frac{105 - 4(17.5) - 0}{8}$$

$$= 4.375$$

$$z_1 = \frac{65 - 5x_1 - 4y_1}{-10} = 4$$

Second iteration,

$$x_2 = \frac{105 - y_1 - z_1}{6} = \frac{105 - 4.375 - 4}{6} = 16.104$$

$$y_2 = \frac{155 - 4x_2 - 3z_1}{8} = 9.823$$

$$z_2 = \frac{65 - 5x_2 - 4y_2}{-10} = \frac{65 - 5(16.104) - 4(9.823}{-10}$$

$$= 5.4812$$

Third iteration,

$$x_3 = \frac{105 - y_2 - z_2}{6} = \frac{105 - 9.823 - 5.4812}{6} = 14.9493$$

$$\approx 15$$

$$y_3 = \frac{155 - 4x_3 - 3z_2}{8} = \frac{155 - 4(14.9493) - 3(5.4812)}{8}$$

$$= 9.8449$$

$$\approx 10$$

$$z_3 = \frac{65 - 5x_3 - 4y_3}{-10} = \frac{65 - 5(14.9493) - 4(9.8449)}{-10}$$

$$= 4.91261$$

$$\approx 5$$

2. Crout's Method:

Ex: Apply crout's method to solve the equations.

$$3x+2y+7z=4$$
$$2x+3y+z=5$$
$$3x+4y+z=7$$

Soln: Given system can be written in the matrix form as

$$\begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix} \longrightarrow (1)$$

$$AX = B$$

Let,

$$A = LU = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12}+l_{22} & l_{21}u_{13}+l_{22}u_{23} \\ l_{31} & l_{31}u_{12}+l_{32} & l_{31}u_{13}+l_{32}u_{23}+l_{33} \end{bmatrix}$$

$\therefore$ $l_{11}=3$; $l_{21}=2$; $l_{31}=3$

$$l_{11}\cdot u_{12}=2 \Rightarrow u_{12}=\frac{2}{l_{11}}=\frac{2}{3}$$

$$l_{11}\cdot u_{13}=7 \Rightarrow u_{13}=\frac{7}{l_{11}}=\frac{7}{3}$$

$$l_{21}u_{12}+l_{22}=3$$
$$l_{22}=3-l_{21}u_{12}$$
$$=3-2\cdot\frac{2}{3}$$
$$=3-\frac{4}{3}=5/3$$

$$l_{21} \cdot u_{13} + l_{22}\, u_{23} = 1$$

$$2 \cdot \frac{7}{3} + \frac{5}{3}\, u_{23} = 1$$

$$\frac{5}{3}\, u_{23} = 1 - \frac{14}{3}$$

$$u_{23} = \frac{-11}{3} \cdot \frac{3}{5}$$

$$u_{23} = \frac{-11}{5}$$

$$l_{31}\, u_{12} + l_{32} = 4$$

$$\cancel{3} \cdot \frac{2}{\cancel{3}} + l_{32} = 4$$

$$l_{32} = 4 - 2$$

$$l_{32} = 2$$

$$l_{31}\, u_{13} + l_{32}\, u_{23} + l_{33} = 1$$

$$\cancel{3} \cdot \frac{7}{\cancel{3}} + 2 \cdot \left(\frac{-11}{5}\right) + l_{33} = 1$$

$$7 - \frac{22}{5} + l_{33} = 1$$

$$\frac{13}{5} + l_{33} = 1$$

$$l_{33} = 1 - \frac{13}{5}$$

$$l_{33} = \frac{-8}{5}$$

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 5/3 & 0 \\ 3 & 2 & -8/5 \end{bmatrix} \quad \& \quad U = \begin{bmatrix} 1 & 2/3 & 7/3 \\ 0 & 1 & -11/5 \\ 0 & 0 & 1 \end{bmatrix}$$

from eqn (1) $\Rightarrow$

$$LUX = B$$

let,

$$UX = V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \longrightarrow (2)$$

from (2), $LV = B$ $\longrightarrow (3)$

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & 5/3 & 0 \\ 3 & 2 & -8/5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$3v_1 = 4 \Rightarrow v_1 = 4/3$

$2v_1 + \frac{5}{3} v_2 = 5 \Rightarrow \frac{5}{3} v_2 = 5 - 2v_1 = 5 - 2\cdot\frac{4}{3}$

$= 5 - \frac{8}{3}$

$v_2 = \frac{7}{3} \cdot \frac{3}{5}$

$\boxed{v_2 = 7/5}$

$3v_1 + 2v_2 - \frac{8}{5} v_3 = 7$

$$\frac{8}{5} v_3 = 3v_1 + 2v_2 - 7$$

$$= 3 \cdot \frac{4}{3} + 2 \cdot \frac{7}{5} - 7$$

$$= 4 + \frac{14}{5} - 7$$

$$= -3 + \frac{14}{5}$$

$$\frac{8}{5} v_3 = \frac{-1}{5}$$

$$v_3 = -\frac{1}{5} \cdot \frac{5}{8}$$

$$\boxed{v_3 = -\frac{1}{8}}$$

eqn(2) $\Rightarrow$ $UX = V$

$$\begin{bmatrix} 1 & 2/3 & 7/3 \\ 0 & 1 & -11/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4/3 \\ 7/5 \\ -1/8 \end{bmatrix}$$

$$x + \frac{2}{3}y + \frac{7}{3}z = \frac{4}{3} \quad \text{——— (1)}$$

$$y - \frac{11}{5}z = \frac{7}{5} \quad \text{——— (2)}$$

$$\boxed{z = -\frac{1}{8}} \quad \text{——— (3)}$$

eqn (2) $\Rightarrow$ $y = \frac{7}{5} + \frac{11}{5}z = \frac{7}{5} + \frac{11}{5}\cdot\left(\frac{-1}{8}\right)$

$$= \frac{7}{5} - \frac{11}{5\cdot 8}$$

$$= \frac{1}{5}\left(7 - \frac{11}{8}\right)$$

$$= \frac{1}{\cancel{5}}\left(\frac{\cancel{45}^{9}}{8}\right)$$

$$\boxed{y = \frac{9}{8}}$$

$$\cancel{\frac{7}{3}z = \frac{4}{3} - x - \frac{2}{3}y}$$

$$\cancel{= \frac{4}{3} -}$$

eqn (3) $\Rightarrow$ $x = \frac{4}{3} - \frac{2}{3}y - \frac{7}{3}z = \frac{4}{3} - \frac{\cancel{2}}{\cancel{3}}\left(\frac{\cancel{9}^{3}}{\cancel{8}_{4}}\right) - \frac{7}{3}\left(\frac{-1}{8}\right)$

$$= \frac{4}{3} - \frac{3}{4} + \frac{7}{24}$$

$$= \frac{32 - 18 + 7}{24} = \frac{\cancel{21}^{7}}{\cancel{24}_{8}} = \frac{7}{8}$$